

SNAPFRAME · DESIGN DECISION RECORD

SnapFrame recommendations accepted

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	SNF-DDR-002	Version	0.2
Status	Draft	Date	2026-09-26
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/snapframe		
Source	docs/decisions/0002-recommendations-accepted.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-26	Amish Chadha	Budget set to \$445 to cover the priced BOM: decided by Amish, 2026-09-26 (O4 closed)
0.1	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); record the newly decided items, what changed and the items still open

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted (items D8 and D9, and the budget under O4); items O1 to O3 and O5 to O8 remain proposed

■ Context

The TRL 3 session (docs/REVIEW.md , "Session 2026-09-25: TRL 3") left five new items awaiting Amish, numbered 4 to 8 in that note, alongside the three open items O1 to O3 of SNF-DDR-001. On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." Every item that carried a recommendation is therefore decided as recommended. Items without a recommendation stay "Proposed, awaiting Amish". TRL 4 remains on hold by Amish's instruction, and nothing here goes past TRL 3.

■ Options considered

The options are in docs/REVIEW.md (TRL 3 session, "Still awaiting Amish") and in SNF-CAL-001 v0.1, Table 5. They are summarized with each item below.

■ Decision

Table 1. Newly decided items.

#	Item (REVIEW.md)	Options	Decision	What changed in the repo
D8	R6 wind (item 4)	(a) 1 in ridge tubes (+2 sticks of 1 in, about \$10; 19.7 m/s); (b) 1 in ridge tubes and posts (about \$40 more; 20.3 m/s); (c) keep 3/4 in ridge tubes and rate the kit at 17.5 m/s	Decided by Amish, 2026-09-25: go with recommendation. (a) 1 in EMT ridge tubes now; a frame analysis comes before any decision on 1 in posts	TUBE_OF["ridge"] set to 1 in in cad/src/model.py ; ridge node ridge bores 24.0 to 30.1 mm; STEP and STL re-exported; BOM item 3 from 3/4 in at \$9.00 to 1 in at \$14.00; SNF-CAL-001 v0.2: rating 17.8 to 19.7 m/s, open-gable case 13.3 to 14.7 m/s, posts now govern (factor 1.46); SNF-DWG-001 Rev P1 to P2
D9	R12 node variants (item 7)	Use the four-socket middle eave node at the corners with one blank socket, removing the handed corner nodes	Decided by Amish, 2026-09-25: go with recommendation. One eave node variant at all six eave positions; the corner socket past the gable stays blank under a push-in cap	node_variants() returns four variants per size (foot, eave, ridge end, ridge middle) instead of six; the handed eave-corner-L and eave-corner-R STL files were removed; BOM items 6 and 12 updated (four caps added); R12 from at risk to met

Consequences of D8 and D9 together, size M (SNF-CAL-001 v0.1 to v0.2):

- Frame kit cost \$431.09 to \$443.30 (7.8 % to 10.8 % over the \$400 budget_usd); budget_usd stays 400 (SNF-DDR-001 D1).
- Frame kit mass 40.6 to 42.0 kg; with tarpaulins 49.8 to 51.1 kg, so **R11 changes from met to not met** (1.1 kg over the 50 kg carry limit).

- Tube bundle 27.4 to 28.6 kg (R5 still not met, now 3.6 kg over).
- Printed nodes 4.87 kg and \$122.09 to 4.97 kg and \$124.30.
- Requirement status: 4 met, 5 not met, 3 at risk, 1 not verifiable, to 4 met, 6 not met, 2 at risk, 1 not verifiable.

Documents revised: SNF-PRC-001 v0.4, SNF-REQ-001 v0.4, SNF-CAL-001 v0.2, SNF-DWG-001 Rev P2, `bom/`
`bom.csv` and `bom/bom-notes.md`, `media/` regenerated. `pitch`, `problem` and `budget_usd` in `project.yaml` are
unchanged, because no accepted recommendation touched them.

Table 2. Items that remain open.

#	Item	Why it is open	Status
O1	Tube offcuts (39 %)	The recommendation was to decide once the first region's tube source is known (SNF-DDR-001)	Proposed, awaiting Amish
O2	First co-design partner and region	No recommendation was made	Proposed, awaiting Amish
O3	Open front gable (R7)	No recommendation was made	Proposed, awaiting Amish
O4	R10 cost (\$443): bulk EMT pricing, a lighter foot plate, or raise <code>budget_usd</code>	Options only, no recommendation	Decided by Amish, 2026-09-26: budget set to \$445 (see below)
O5	R5 tube bundle (28.6 kg) and, now, R11 carried mass (51.1 kg): two tube bundles (16.5 and 12.1 kg), or relax R5 to 30 kg for a two-person carry; R11 also needs a choice	Options only, no recommendation	Proposed, awaiting Amish
O6	Node polymer (R9)	No recommendation was made	Proposed, awaiting Amish
O7	R8 anchor target: 1.0 kN target against 1.28 kN calculated demand	No recommendation was made	Proposed, awaiting Amish
O8	1 in posts (R6 would reach 20.3 m/s by member check)	D8 puts this after a frame analysis, which has not been run yet	Proposed, awaiting Amish once the frame analysis is done

Budget, 2026-09-26. On 2026-09-26 Amish wrote: "i approve all the budget items." Budget set to \$445 to cover the priced BOM: decided by Amish, 2026-09-26. The priced frame kit is \$443.32 (`bom/bom.csv` , excluding the agency-stock tarpaulins under SNF-DDR-001 D1), so `budget_usd` in `project.yaml` moves from 400 to 445 and R10 moves from not met to met. SNF-CAL-001 v0.3, SNF-REQ-001 v0.5, SNF-PRC-001 v0.5, SNF-PRB-001 v0.6, `README.md` , `bom/bom-notes.md` and the concept blueprint were updated. Requirement status is now 5 met, 5 not met, 2 at risk and 1 not verifiable.

■ Consequences

- R6 remains not met at 19.7 m/s. The frame analysis named in D8 is TRL 3 paper work and is the next calculation step; it was not run in this session.
- R11 is newly not met because the 1 in ridge tubes add 1.2 kg. This is reported to Amish under O5 and is not decided here.
- No cross-repo action follows from these decisions.

- TRL 4 work (built frame, node and anchor load tests, timed erection, purchasing) remains on hold by Amish's instruction.